CAMPUSLINK

The Career Intelligence Operating System for Higher Education

Bridging the campus-to-corporate credibility gap through explainable matching, rubric-verified skills, and zero black-box AI.

Target Audience: Colleges, Students, Placement Cells, Recruiters	**Status:** Fully Functional Local-First Architecture
Core Innovation: Deterministic Rubric & Verified Provenance	**Tech Stack:** React 19, TypeScript, Express, PostgreSQL, Prisma

1. Problem Statement & The CampusLink Solution

Why traditional placements and black-box hiring platforms are failing today

THE PROBLEM IN CAMPUS HIRING

Resume Credibility Inflation

Students write buzzwords and self-report 'expert' skills without verifiable evidence, forcing recruiters to doubt every resume.

Opaque 'Black-Box' AI Matchers

Existing recruitment platforms use unexplainable ML classifiers. Rejected candidates never know why they were cut or how to improve.

Manual, Disconnected Placement Cells

Colleges manage drives across WhatsApp groups, Google Sheets, and email blasts with zero verifiable audit trails.

THE CAMPUSLINK TRANSFORMATION

Provenance-Based Skill Passport

Separates self-reported claims from rubric-assessed proofs. Unverified claims earn zero evidence points.

100% Explainable Matching

Deterministic mathematical formula breaks down exact met criteria, missing gaps, bonus points, and next actionable steps.

Unified Placement Tower OS

Real-time cohort readiness analytics, guarded application state machine, and frozen materialized announcement broadcasts.

2. How Things Work: The 5-Step Student Journey

From initial onboarding to verified readiness and job placement

1. Career Twin

Students record skills across 4 standardized tiers (Beginner to Expert). Unverified claims are flagged until proven by project evidence.

2. Opportunity Mirror

Algorithm computes match score (0-100%) against corporate job criteria with full transparency into met criteria and skill gaps.

3. Skill Sprint

Students solve coding challenges. Submissions are scored via Rubric v1. Scoring 70+ auto-promotes skill and records project proof.

4. Interview Arena

Mock interview practice room scoring STAR structure, substance, and quantitative metrics with an anti-duplication points ledger.

5. Verified Passport

Holistic Readiness Score (50% Proficiency + 30% Evidence + 20% Momentum) unlocks guarded corporate applications.

CORE PHILOSOPHY: ZERO BLACK-BOX AI

Every assessment score and match calculation uses published, deterministic formulas. Candidates receive actionable feedback on exactly which engineering decisions earned or lost points, fostering true pedagogical improvement rather than arbitrary algorithmic rejection.

3. Transparent Mathematical Engines

Pure deterministic logic: auditable, explainable, and tamper-resistant

Explainable Match Formula (lib/match.ts)

- Weight = Rank(minLevel) x 2
- Met: Full Weight + min(Projects, 2) x 0.5
- Partial: 75% weight for 1 level below
- Nice-to-Have: +1 point at Intermediate+
- Score = min(100, round(100 x Pts / Max))

Sprint Rubric v1 (lib/rubric.ts)

- 40 Base Points
- +25 Full Checklist Completed
- +15 Verifiable Repo / Live Deploy URL
- +15 Write-up Depth (>=40 words)
- +5 Quality Attributes (tests, perf, a11y)
- Threshold: 70+ auto-promotes skill tier

Holistic Student Readiness

- 50% Proficiency: (Mean Skill Rank / 4) x 50
- 30% Evidence: (Evidenced / Total) x 30
- 20% Momentum: min(20, Apps x 4 + Sprints x 6)
- Unverified claims receive zero evidence points
- Direct pedagogical incentive for real code

4. System Architecture & Complete Technology Stack

Local-first monorepo built for enterprise security and resilience

Frontend Layer

- React 19 & Vite 8 (Strict port 5173)
- React Router v7 & Session Gates
- TanStack Query (15s cache, auto-sync)
- Tailwind CSS v4 & Lucide Icons

API & Backend Layer

- Express 4.21 & TypeScript 5.6
- Zod strict runtime schema parsing
- Helmet, dynamic CORS & CSRF guard
- Multi-tier Express-Rate-Limit budgets

Database & Security

- PostgreSQL 16+ & Prisma ORM 5
- scrypt password hashing (memory-hard)
- CSPRNG HMAC-SHA256 Email OTP
- Revocable HttpOnly session tokens

DEFENSE-IN-DEPTH POSTURE

Bound strictly to loopback 127.0.0.1; no wildcard interface exposure. Server-side RBAC enforces permissions strictly: client-submitted studentProfileId or companyId parameters are ignored in favor of the authoritative server session. Zero database errors or stack traces leak to clients.

5. What Is Built & Grand Finale Deliverables

From current working prototype to final competition launch

ALREADY BUILT & VERIFIED TODAY (100% WORKING)

- 3-Step Email-verified student registration with live SMTP OTP
- Career Twin with 4 skill tiers and linked project evidence
- Opportunity Mirror with explainable match breakdown
- Skill Sprint with live Rubric v1 grading & auto-promotion
- Interview Arena with STAR evaluation & point ledgers
- Placement Tower with cohort readiness & What-If Lab
- Guarded Application FSM (Applied -> Shortlisted -> Hired)

COMMITMENTS FOR GRAND FINALE LAUNCH

- Multi-College Placement Consortium: Cross-campus placement drives
- Cryptographically Signed PDF Skill Passports (W3C Verifiable Credentials)
- Recruiter Bulk Resume Screening Portal with deterministic scoring
- Automated Calendar Integration for shortlisted interview slots
- Dockerized 1-click self-hosted deployment for university IT cells

6. The Vision: Restoring Trust in Campus Placements

Empowering millions of students with verifiable career trajectory

For Students

Replaces resume anxiety with a transparent growth roadmap. Every sprint closed directly lifts job match scores and readiness percentages.

For Placement Cells

Replaces chaotic spreadsheets with executive cohort intelligence, verified audit trails, and targeted student interventions.

For Corporate Recruiters

Cuts screening time by 80% by filtering candidates on verified code proofs rather than unprovable self-attested claims.

CONCLUSION

CAMPUSLINK transforms campus placements from an opaque lottery into a merit-based, evidence-driven career intelligence ecosystem. Built with production-grade engineering, local-first reliability, and mathematically auditable integrity.